Transformer-based approach for Music Identification

CS7150 Final Project Concept Presentation

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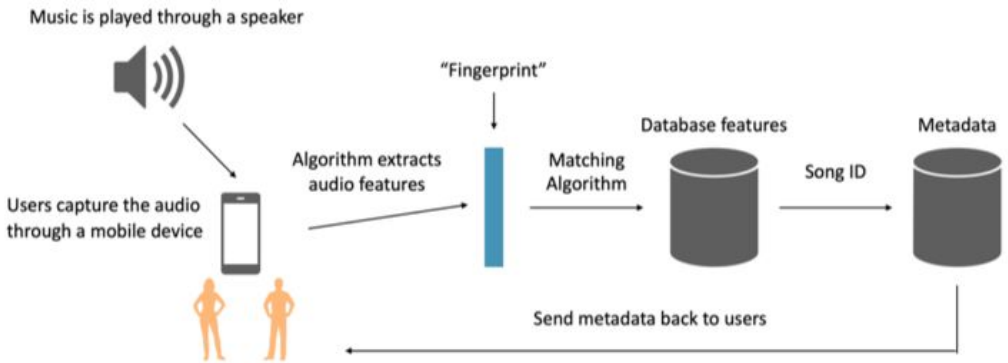
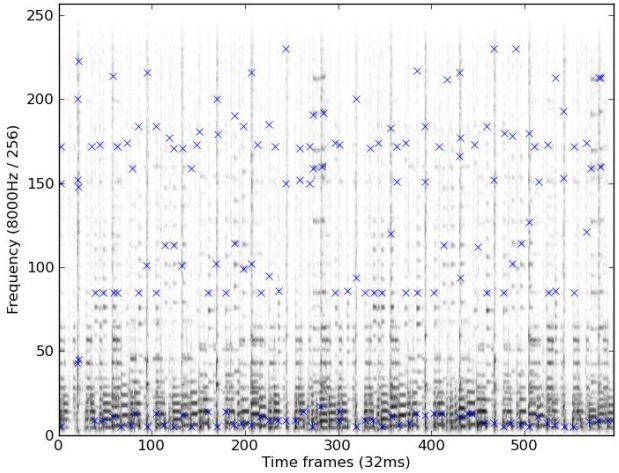
Music Identification & Analysis Approaches

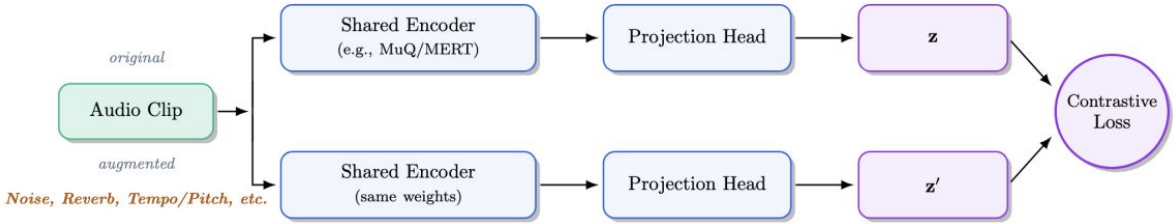
- Existing Approaches:
  - Fingerprinting (Shazam)
    - Lacks learning and generalization
  - CNN
    - Learns local patterns but still misses the global context





Project Idea & Methods

- Idea
  - Song identification with a transformer-based approach
  - Compare with existing solutions
  - Apply noise and small offsets to simulate real-world situations
- Methods
  - Transformer model
  - Contrastive learning



Datasets and Models

- Datasets
  - Slakh2100 (http://www.slakh.com/)
  - MUSDB18(https://sigsep.github.io/datasets/musdb.html)
  - FMA (https://arxiv.org/pdf/1612.01840)
- Existing benchmarks
  - Pexeso (https://github.com/Pexeso/audio-fingerprinting-benchmark-toolkit)

Deliverables expected for the project

- Data processing pipeline to process raw audio into spectrograms
- A transformer model that can match audio clips to the original songs (and other top matches)
- Compare with other existing approaches (shazam or CNN)
- Testing the effects with noise in the background to simulate real-world scenarios

Thank you!

Any questions?